

Udayan Atreya

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EDUCATION

Master of Science, Computer Science

Expected May 2026

San Jose State University, San Jose, US

Coursework: Retrieval Augmented Generation (RAG), Applied Deep Learning, Object-Oriented Design.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures & Algorithms, Relational Database Management Systems, OOPS.

TECHNICAL SKILLS

Languages: Python, Java, C#, SQL, HTML, CSS, JavaScript

Frameworks & APIs: FastAPI, REST APIs, LangChain, RAG, HuggingFace, TensorFlow, PyTorch, Git, .NET

Cloud & Database: AWS (API Gateway, Lambda, S3), CI/CD, MongoDB, DynamoDB, Pinecone, ChromaDB

LLMs & AI Tools: Claude, GPT, Gemini, Ollama, Llama, DeepSeek, Groq, Vapi, Eleven Labs

PROJECT EXPERIENCE

Agentic BMC Observability Platform (*FastAPI, LangChain, Gemini 1.5, MongoDB, S3*)

[\[GitHub\]](#)

Jun 2025 – Jul 2025

- Built a real-time web observability platform featuring **Grafana and Prometheus web dashboards** backed by **RESTful FastAPI endpoints**, enabling seamless frontend–backend data exchange and achieving **95% query classification accuracy** while **reducing triage time by 30%**.
- Integrated a hybrid retrieval backend (MongoDB, S3) powering live dashboard data feeds, **cutting retrieval latency by 75%** (2m to 30s) for accelerated root cause analysis.
- Enforced role-based access control and human-in-the-loop API validation on hardware interventions, **preventing 100% of unauthorized state changes** across Baseboard Management Controllers.

Evaluator-Optimizer RAG (*LangChain, GPT-4o, Gemini 3 Pro, Llama-3, ChromaDB*)

[\[GitHub\]](#)

Aug 2025 – Mar 2026

- Architected an agentic RAG framework exposing clean, well-documented **REST API interfaces**, **reducing hallucination by 25%** and **improving answer accuracy by 18%** over baselines for Language-Based Video Reasoning systems.
- Engineered a **RAG evaluation suite** using RAGAS and Llama-3-70b (LLM-as-a-judge) across 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Context Precision, and Semantic Similarity.
- Optimized latency-accuracy trade-off via Dynamic Chunking and MMR search, **reducing context redundancy by 40%** while maintaining **90%+ semantic coverage** across test corpora.

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Architected a user-facing notification service with **RESTful APIs (API Gateway, Lambda, DynamoDB)**, handling frontend–backend data exchange for remediation workflows and driving a **20% increase in DAU** post-launch.
- Optimized **backend API performance** by redesigning DynamoDB access patterns (schema, pk, sk) and CRUD microservices, **reducing retrieval latency by 37%** for high-traffic web workloads.
- Delivered a code vulnerability remediation solution via serverless microservices (AWS Lambda, S3, EventBridge) and Azure DevOps CI/CD pipelines, safeguarding **400+ enterprise applications** across orgs.
- Designed a single-touch server patching solution integrated with **ServiceNow web workflows** using serverless architecture (AWS, Azure DevOps), **reducing MTTR by 80%**.
- Partnered with **cross-functional teams including design** to ship a **ChatOps web platform** routing natural language queries to automation workflows via integrated APIs, **reducing overhead by 40%**.
- Built an analytics-driven recommendation engine for 100+ automation assets, implementing weighted scoring to surface personalized tool suggestions for users, **reducing discovery time by 30%**.